# Solutions to Exercises 2

**Exercise 1** (Traces)**.**

1. Since there are $n$ variables, which can take two values, there are $2^n$ possible states. We can construct a trace passing through each state exactly once.

2. As in class, $\bar{s} = (s_1, \dots s_n)$ and exponents, e.g. $\bar{s}^k$, are used to represent variables in different states (i.e. $\bar{s}^k = (s_1^k, \dots s_n^k)$).

   (a) Noting
   $$\bar{v}^k = (s_1^k, \dots, s_n^k, a_1^k, \dots, a_m^k, x_1^k, \dots, x_\ell^k),$$
   $$F_{(=k)} = \exists \bar{v}^0, \dots, \bar{v}^{k-1}, \bar{s}^k.\ \mathit{Init}\left[\bar{s} := \bar{s}^0\right] \wedge \left(\bigwedge_{i=0}^{k-1} R_i\right) \wedge \left(\bar{s}^k = \bar{s}\right).$$

   Another (equivalent) solution:

   $\exists \bar{v}^0, \dots, \bar{v}^{k-1}.$
   $$\mathit{Init}\left[\bar{s} := \bar{s}^0\right] \wedge \left(\bigwedge_{i=0}^{k-2} R_i\right) \wedge R\left[(\bar{s}, \bar{a}, \bar{x}, \bar{s}') := (\bar{s}^{k-1}, \bar{a}^{k-1}, \bar{x}^{k-1}, \bar{s})\right].$$

   (b) $F_{(\leq k)} = \bigvee_{i=0}^{k} F_{(=i)}$.

   (c) The "intuitive" solution is $F_{(\leq \infty)}$. However, this formula is infinite. Using question (1), it can be reduced to $F_{(\text{reachable})} = F_{(\leq 2^n - 1)}$.

   (d) $F_{(=k,\text{Err})} = \exists \bar{v}^0, \dots, \bar{v}^{k-1}, \bar{s}^k.\ \left(\bar{s}^0 = \bar{s}\right) \wedge \left(\bigwedge_{i=0}^{k-1} R_i\right) \wedge E\left[\bar{s} := \bar{s}^k\right]$

   (e) $F_{(\leq k,\text{Err})} = \bigvee_{i=0}^{k} F_{(=i,\text{Err})}$

   (f) $F_{(\text{stuck})} = \neg \exists \bar{a}, \bar{x}, \bar{s}'.\ R$

3. The formulas are:

   (a) $\neg \exists \bar{s}.\ F_{(\text{reachable})} \wedge E$

   (b) $\neg \exists \bar{s}.\ F_{(\text{reachable})} \wedge F_{(\text{stuck})}$

   (c) $\forall \bar{s}.\ F_{(\leq k)}$

(d) Note that we are only allowed to quantify on propositional variables. As such, a formula such as

$$\forall k.\ \exists \bar{s}.\ F_{(=k)} \land \neg F_{(\text{stuck})}$$

is not allowed, as $k$ ranges (implicitly) over $\mathbb{N}$.

Consider instead

$$\exists \bar{s}.\ F_{(=2^n)}$$

if such a state ($\bar{s}$) exists, the underlying trace contains $2^n + 1$ states (since it is $2^n$ steps long). By the pigeon-hole principle, one of these states must appear twice in the trace. This means that there is a part of the trace which can be repeated infinitely ("pumped"), implying the existence of an infinite trace (see the proof of the pumping lemma for regular languages which relies on the same principle).

**Exercise 2** (Tseytin Transformation)**.**

1. Any expression of the type $y_k \leftrightarrow x_i * x_j$ is easily convertible to a CNF.

$$\begin{aligned}
y_k \leftrightarrow x_i \lor x_j &\equiv (y_k \to x_i \lor x_j) \land (x_i \lor x_j \to y_k) \\
&\equiv (\neg y_k \lor x_i \lor x_j) \land (\neg x_i \lor y_k) \land (\neg x_j \lor y_k) \\
y_k \leftrightarrow x_i \land x_j &\equiv (y_k \to x_i \land x_j) \land (x_i \land x_j \to y_k) \\
&\equiv (\neg y_k \lor x_i) \land (\neg y_k \lor x_j) \land (\neg x_i \lor \neg x_j \lor y_k) \\
y_k \leftrightarrow x_i \to x_j &\equiv y_k \leftrightarrow \neg x_i \lor x_j \\
&\equiv (\neg y_k \lor \neg x_i \lor x_j) \land (x_i \lor y_k) \land (\neg x_j \lor y_k) \\
y_k \leftrightarrow \neg x_i &\equiv (y_k \to \neg x_i) \land (\neg x_i \to y_k) \\
&\equiv (\neg y_k \lor \neg x_i) \land (x_i \lor y_k)
\end{aligned}$$

Let's first define the mapping between sub formulas and fresh variables starting from the leaves:

$$\begin{aligned}
&y_1 \leftrightarrow \neg b \\
&y_2 \leftrightarrow a \lor y_1 \\
&y_3 \leftrightarrow b \to c \\
&y_4 \leftrightarrow \neg y_3 \\
&y_5 \leftrightarrow y_2 \land y_4 \\
&y_6 \leftrightarrow y_5 \to a \\
&y_7 \leftrightarrow \neg y_6
\end{aligned}$$

The Tseytin transformation of the formula is therefore:

$$\begin{aligned}
&(y_1 \leftrightarrow \neg b) \wedge (y_2 \leftrightarrow a \vee y_1) \wedge (y_3 \leftrightarrow b \rightarrow c) \wedge (y_4 \leftrightarrow \neg y_3) \wedge \\
&(y_5 \leftrightarrow y_2 \wedge y_4) \wedge (y_6 \leftrightarrow y_5 \rightarrow a) \wedge (y_7 \leftrightarrow \neg y_6) \wedge y_7 \\
=&(\neg y_1 \vee \neg b) \wedge (b \vee y_1) \wedge (\neg y_2 \vee a \vee y_1) \wedge (\neg a \vee y_2) \wedge (\neg y_1 \vee y_2) \wedge \\
&(\neg y_3 \vee \neg b \vee c) \wedge (b \vee y_3) \wedge (\neg c \vee y_3) \wedge (\neg y_4 \vee \neg y_3) \wedge (y_3 \vee y_4) \wedge \\
&(\neg y_5 \vee y_2) \wedge (\neg y_5 \vee y_4) \wedge (\neg y_2 \vee \neg y_4 \vee y_5) \wedge (\neg y_6 \vee \neg y_5 \vee a) \wedge \\
&(y_5 \vee y_6) \wedge (\neg a \vee y_6) \wedge (\neg y_6 \vee \neg y_7) \wedge (y_6 \vee y_7) \wedge y_7
\end{aligned}$$

2. We first show that they might not be equivalent by providing a small counterexample. As written above, the Tseytin transform of the formula $\neg x$ is $(y \leftrightarrow \neg x) \wedge y$. The value of $\neg x$ when $x = 0$ is 1, whereas the value of its Tseytin transformation when $x = 0, y = 0$, is 0. Therefore the two formulas are not equivalent.

   Let's now prove that a formula is satisfiable if and only if its Tseytin transformation is satisfiable as well.

   Let $\varphi$ be a formula and $V = \mathrm{FV}(\varphi)$ its set of free variables. One can notice (and prove by induction) that the Tseytin transformation of $\varphi$ is equivalent to:

$$y_n \wedge \bigwedge_{i=1}^{n} (y_i \leftrightarrow \psi_i)$$

   where $\{\psi_i\}_{0 \leq i \leq n}$ are all the (non-atomic) subformulas of $\varphi$; in particular, $\psi_n = \varphi$.

   $\varphi$ is satisfiable iff there exists an assignment $\alpha$ such that $\varphi\,[V := \alpha] = 1$. However, note that the formula $(y_i \leftrightarrow \psi_i)$ is true if and only if one assigns the truth value of $\psi_i\,[V := \alpha]$ to $y_i$. We can construct a new assignment $\alpha'$ as follows:

$$\alpha'(v) = \begin{cases} \psi_i\,[V := \alpha] & \text{if } v = y_i \text{ for some } 0 \leq i \leq n \\ \alpha(v) & \text{otherwise} \end{cases}$$

   Convince yourself that this is a satisfying assignment for both the original formula $\varphi$ and its Tseytin transformation. Conversely, let $\alpha'$ be a satisfying assignment for the Tseytin transformation of $\varphi$. This is already a satisfying assignment for $\varphi$.

   Thus, there exists a satisfying assignment for $\varphi$ if and only if there exists a satisfying assignment for its Tseytin transformation. Therefore, they are equisatisfiable.

3. $F$ is a valid formula if and only if $\neg F$ is false for any variable assignment, i.e. it is unsatisfiable. A formula is unsatisfiable if and only if its Tseytin transformation is unsatisfiable as well (by the previous question). Therefore $F$ is a valid formula if and only if the Tseytin transformation of $\neg F$ is unsatisfiable.

By induction on the structure of the formula, one can show that formula size increases by at most a constant amount for each "building block" of the formula (literal or binary operator). Therefore the size of the Tseytin transformation is in $O(n)$.

**Exercise 3** (SAT solving)**.** By taking the CNF found in the above exercise and applying resolution, we obtain the following derivation trees:

$$\dfrac{\neg a \lor y_6 \qquad \dfrac{y_7 \qquad \neg y_6 \lor \neg y_7}{\neg y_6}}{\neg a \ (1)} \qquad\qquad \dfrac{b \lor y_3 \qquad \dfrac{\neg y_3 \lor \neg y_4 \qquad \dfrac{y_4 \lor \neg y_5 \qquad \dfrac{y_5 \lor y_6 \qquad \dfrac{y_7 \qquad \neg y_6 \lor \neg y_7}{\neg y_6}}{y_5}}{y_4}}{\neg y_3}}{b \ (2)}$$

$$\dfrac{\dfrac{\neg y_1 \lor \neg b \qquad \neg y_2 \lor a \lor y_1}{\neg b \lor a \lor \neg y_2} \qquad \dfrac{\neg y_5 \lor y_2 \qquad \dfrac{y_5 \lor y_6 \qquad \dfrac{y_7 \qquad \neg y_6 \lor \neg y_7}{\neg y_6}}{y_5}}{y_2}}{\neg b \lor a \ (3)} \qquad\qquad \dfrac{b \ (2) \qquad \dfrac{\neg b \lor a \ (3) \qquad \neg a \ (1)}{\neg b}}{F}$$

**Exercise 4** (First-order formulas)**.** The proofs are not given for all cases. The omitted cases are similar to those already given.

1. $A : \forall x.(P(x) \lor Q(x))$
   $B : (\forall x.P(x)) \lor (\forall x.Q(x))$

   (a) $A \models B$: ✗. Consider $(\mathbb{N}, e)$ with

   $$e(P) = \{n \in \mathbb{N} \mid n \bmod 2 = 0\}$$
   $$e(Q) = \{n \in \mathbb{N} \mid n \bmod 2 = 1\}$$

   (b) $B \models A$: ✓. Consider any structure $(D, e)$ such that $(D, e) \models B$. We have $[\![B]\!]_{(D,e)} = 1$. By definition of $[\![\cdot \lor \cdot]\!]_{(D,e)}$, we know that either $[\![\forall x.P(x)]\!]_{(D,e)} = 1$ or $[\![\forall x.Q(x)]\!]_{(D,e)} = 1$. Pick any case, the other is identical. Supposing $[\![\forall x.P(x)]\!]_{(D,e)} = 1$, we know that for every $d \in D$, $[\![P(x)]\!]_{(D,e[x:=d])} = 1$. However, this implies that for every $d \in D$, $[\![P(x) \lor Q(x)]\!]_{(D,e[x:=d])} = 1$, and thus $[\![\forall x.(P(x) \lor Q(x))]\!]_{(D,e)} = 1$. Therefore, $(D, e) \models A$ as well.

2. $A : \exists x.(P(x) \lor Q(x))$
   $B : (\exists x.P(x)) \lor (\exists x.Q(x))$

   (a) $A \models B$: ✓. Consider any structure $(D, e)$ such that $(D, e) \models A$. We have $[\![A]\!]_{(D,e)} = 1$. By definition of $[\![\cdot]\!]_{(D,e)}$, there exists $d \in D$ such that $[\![P(x) \lor Q(x)]\!]_{(D,e[x:=d])} = 1$. Thus, either $[\![P(x)]\!]_{(D,e[x:=d])} = 1$ or $[\![Q(x)]\!]_{(D,e[x:=d])} = 1$. In the first case, $[\![\exists x.P(x)]\!]_{(D,e)} = 1$, and in the second case, $[\![\exists x.Q(x)]\!]_{(D,e)} = 1$. In either case, $(D, e) \models B$.

(b) $B \models A$: ✓. Proof omitted.

3. $A : \forall x.(P(x) \land Q(x))$
   $B : (\forall x.P(x)) \land (\forall x.Q(x))$

   (a) $A \models B$: ✓. Proof omitted.

   (b) $B \models A$: ✓. Proof omitted.

4. $A : \exists x.(P(x) \land Q(x))$
   $B : (\exists x.P(x)) \land (\exists x.Q(x))$

   (a) $A \models B$: ✓. Proof omitted.

   (b) $B \models A$: ✗. Consider $(\mathbb{N}, e)$ with

$$e(P) = \{0\}$$
$$e(Q) = \{1\}$$

5. $A : \forall x.(P(x) \to Q(x))$
   $B : (\forall x.P(x)) \to (\forall x.Q(x))$

   (a) $A \models B$: ✓. Consider any structure $(D, e)$ such that $(D, e) \models A$. We have $[\![A]\!]_{(D,e)} = 1$, i.e., for every $d \in D$, $[\![P(x) \to Q(x)]\!]_{(D,e[x:=d])} = 1$. Note that $[\![P(x) \to Q(x)]\!]_{(D,e[x:=d])} = [\![\neg P(x) \lor Q(x)]\!]_{(D,e)}$, and thus for each $d$:

$$[\![P(x)]\!]_{(D,e)} = 0 \text{ or } [\![Q(x)]\!]_{(D,e)} = 1 \tag{1}$$

   Assume, towards a contradiction, that $[\![B]\!]_{(D,e)} = 0$. We must have

$$[\![\forall x.P(x)]\!]_{(D,e)} = 1 \text{ and } [\![\forall x.Q(x)]\!]_{(D,e)} = 0 \tag{2}$$

   The left conjunct of Equation 2 tells us that for every $d \in D$, $[\![P(x)]\!]_{(D,e[x:=d])} = 1$. By Equation 1, this implies that for every $d \in D$, $[\![Q(x)]\!]_{(D,e[x:=d])} = 1$. This contradicts the right conjunct of Equation 2. Therefore, $[\![B]\!]_{(D,e)} = 1$, and thus $(D, e) \models B$.

   (b) $B \models A$: ✗. Consider $(\mathbb{N}, e)$ with

$$e(P) = \{0\}$$
$$e(Q) = \{1\}$$

6. $A : \exists x.(P(x) \to Q(x))$
   $B : (\exists x.P(x)) \to (\exists x.Q(x))$

(a) $A \models B$: ✗. Consider $(\mathbb{N}, e)$ with

$$e(P) = \{0\}$$
$$e(Q) = \emptyset$$

Take $x = 1$ as a witness for $A$.

(b) $B \models A$: ✓. Consider any structure $(D, e)$ such that $(D, e) \models B$. We have $[\![B]\!]_{(D,e)} = 1$. As before, by expanding the implication, we must have either $[\![\exists x.P(x)]\!]_{(D,e)} = 0$ or $[\![\exists x.Q(x)]\!]_{(D,e)} = 1$.

In the first case, we have that for every $d \in D$, $[\![P(x)]\!]_{(D,e[x:=d])} = 0$. This implies that for every $d \in D$, $[\![P(x) \to Q(x)]\!]_{(D,e[x:=d])} = 1$, and thus in particular $[\![\exists x.P(x) \to Q(x)]\!]_{(D,e)} = 1$.

In the second case, there exists $d \in D$ such that $[\![Q(x)]\!]_{(D,e[x:=d])} = 1$. Applying the interpretation of implication again, we have that $[\![P(x) \to Q(x)]\!]_{(D,e[x:=d])} = 1$, and thus $[\![\exists x.P(x) \to Q(x)]\!]_{(D,e)} = 1$.

Thus, in either case, $(D, e) \models A$.

**Exercise 5** (First-order structures)**.**

1. Let $I = (D, \alpha)$ be an interpretation of $\mathcal{L}$, for which $\alpha(B) = b$ and $\alpha(T) = t$. For notational convenience we will denote $\alpha(L)$ by $L_\alpha$. The consequences of axioms in $S$ are:

$$\begin{aligned}
[\![\mathsf{Min}]\!]_I = 1 &\quad\Longrightarrow L_\alpha(b,b), L_\alpha(b,u) \text{ and } L_\alpha(b,t)\\
[\![\mathsf{Max}]\!]_I = 1 &\quad\Longrightarrow L_\alpha(b,t), L_\alpha(u,t) \text{ and } L_\alpha(t,t)\\
[\![\mathsf{Tot}]\!]_I = 1 &\quad\Longrightarrow L_\alpha(u,u) \text{ (by choosing } x = u \text{ and } y = u)\\
[\![\mathsf{Tra}]\!]_I = 1 &\quad\Longrightarrow (L_\alpha(t,b) \implies L_\alpha(t,u))
\end{aligned}$$

This greatly reduces the number of choices. We can fill out the partial table for $L_\alpha$:

| $L_\alpha$ | b | u | t |
|---|---|---|---|
| b | 1 | 1 | 1 |
| u | ? | 1 | 1 |
| t | ? | ? | 1 |

Finally, based on $\mathsf{Tra}$, $L_\alpha$ is restricted to be either:

*a)*

| $L_\alpha$ | $b$ | $u$ | $t$ |
|---|---|---|---|
| $b$ | 1 | 1 | 1 |
| $u$ | 1 | 1 | 1 |
| $t$ | 1 | 1 | 1 |

*b)*

| $L_\alpha$ | $b$ | $u$ | $t$ |
|---|---|---|---|
| $b$ | 1 | 1 | 1 |
| $u$ | 1 | 1 | 1 |
| $t$ | 0 | 0 | 1 |

*c)*

| $L_\alpha$ | $b$ | $u$ | $t$ |
|---|---|---|---|
| $b$ | 1 | 1 | 1 |
| $u$ | 0 | 1 | 1 |
| $t$ | 0 | 1 | 1 |

*d)*

| $L_\alpha$ | $b$ | $u$ | $t$ |
|---|---|---|---|
| $b$ | 1 | 1 | 1 |
| $u$ | 0 | 1 | 1 |
| $t$ | 0 | 0 | 1 |

All the options satisfy reflexivity and transitivity as a consequence of the axioms. However, only *d)* satisfies the anti symmetry property. Therefore only d) is a partial order.

2.

*a)*

| $E_{\alpha'}$ | $b$ | $u$ | $t$ |
|---|---|---|---|
| $b$ | 1 | 1 | 1 |
| $u$ | 1 | 1 | 1 |
| $t$ | 1 | 1 | 1 |

*b)*

| $E_{\alpha'}$ | $b$ | $u$ | $t$ |
|---|---|---|---|
| $b$ | 1 | 1 | 0 |
| $u$ | 1 | 1 | 0 |
| $t$ | 0 | 0 | 1 |

*c)*

| $E_{\alpha'}$ | $b$ | $u$ | $t$ |
|---|---|---|---|
| $b$ | 1 | 0 | 0 |
| $u$ | 0 | 1 | 1 |
| $t$ | 0 | 1 | 1 |

*d)*

| $E_{\alpha'}$ | $b$ | $u$ | $t$ |
|---|---|---|---|
| $b$ | 1 | 0 | 0 |
| $u$ | 0 | 1 | 0 |
| $t$ | 0 | 0 | 1 |

In all the cases, $E_{\alpha'}$ is an equivalence relation.

3. $E_{\alpha'}$ is always an equivalence relation, whatever is the domain.

Let $D'$ be an arbitrary set and $I' = (D', \alpha')$.

- **Reflexivity**: $[\![\mathsf{Both}]\!]_{I'} = 1 \implies \forall x \in D'. E_{\alpha'}(x,x) \leftrightarrow L_{\alpha'}(x,x)$ (by choosing $y = x$).

  We also know that, $[\![\mathsf{Tot}]\!]_{I'} = 1 \implies \forall x \in D'. \ L_{\alpha'}(x,x)$

  Therefore $\forall x \in D'. \ E_{\alpha'}(x,x)$

- **Symmetry**:

  $$[\![\mathsf{Both}]\!]_{I'} = 1 \implies \forall x \in D'. \ \forall y \in D'. \ E_{\alpha'}(x,y) \leftrightarrow (L_{\alpha'}(x,y) \wedge L_{\alpha'}(y,x))$$

  We also know that for $x, y \in D'$ arbitrary $L_{\alpha'}(x,y) \wedge L_{\alpha'}(y,x) \equiv L_{\alpha'}(y,x) \wedge L_{\alpha'}(x,y)$. Therefore we have that for $\forall x \in D'. \ \forall y \in D'. \ E_{\alpha'}(x,y) \leftrightarrow E_{\alpha'}(y,x)$ is a tautology.

- **Transitivity**: $[\![\mathsf{Both}]\!]_{I'} = 1$ implies that for $x, y, z$ arbitrary in $D'$

  $$E_{\alpha'}(x,y) \leftrightarrow (L_{\alpha'}(x,y) \wedge L_{\alpha'}(y,x))$$

  and

  $$E_{\alpha'}(y,z) \leftrightarrow (L_{\alpha'}(y,z) \wedge L_{\alpha'}(z,y))$$

  Therefore

  $$E_{\alpha'}(x,y) \wedge E_{\alpha'}(y,z) \leftrightarrow (L_{\alpha'}(x,y) \wedge L_{\alpha'}(y,z) \wedge L_{\alpha'}(z,y) \wedge L_{\alpha'}(y,x)$$

  which implies

  $$E_{\alpha'}(x,y) \wedge E_{\alpha'}(y,z) \rightarrow (L_{\alpha'}(x,z) \wedge L_{\alpha'}(z,x))$$

  and finally

  $$E_{\alpha'}(x,y) \wedge E_{\alpha'}(y,z) \rightarrow E_{\alpha'}(x,z)$$

4. One of the solution consists in choosing $D_1 = {D'}/{E_{\alpha'}}$, the set of equivalence classes of $E_{\alpha'}$, $\alpha_1 = (B_1, T_1, L_1)$ where $B_1 = [b]_{E_{\alpha'}}$, $T_1 = [t]_{E_{\alpha'}}$ and

   $$L_1 := \left\{ \left([x]_{E_{\alpha'}}, [y]_{E_{\alpha'}}\right) : L_{\alpha'}(x,y) \text{ where } x, y \in D' \right\}$$

   Reflexivity and transitivity are immediate as they have been proven for $L_{\alpha'}$.

   Antisymmetry comes from the fact that for arbitrary $[x]_{E_{\alpha'}}, [y]_{E_{\alpha'}} \in D_1$

   $$\begin{aligned} & L_1\left([x]_{E_{\alpha'}}[y]_{E_{\alpha'}}\right) \wedge L_1\left([x]_{E_{\alpha'}}, [y]_{E_{\alpha'}}\right) \\ \Longrightarrow\ & L_{\alpha'}(x,y) \wedge L_{\alpha'}(y,x) \\ \Longrightarrow\ & E_{\alpha'}(x,y) \\ \Longrightarrow\ & [x]_{E_{\alpha'}} = [y]_{E_{\alpha'}} \end{aligned}$$

   Therefore $L_1$ is a partial order.